

Filter Reversing

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Given F_s , Q , and ω_0 , we can begin the calculations for any filter with:

$$\omega_0 = 2\pi \frac{f_0}{F_s}$$

$$\alpha = \frac{\sin \omega_0}{2Q}$$

The coefficients from the spec A_1 and A_2 are normalized (we know this because A_0 is not specified) so we know that:

$$A_1 = \frac{a_1}{a_0}$$

$$A_2 = \frac{a_2}{a_0}$$

Stage 1 filter

Cookbook definition of a high-shelf filter

We need an extra variable, A which is our gain. If we have gain in decibels G , it can be converted with $A = \sqrt{10^{G/20}} = 10^{G/40}$.

$$b_0 = A \left((A + 1) + (A - 1) \cos \omega_0 + 2\sqrt{A}\alpha \right)$$

$$b_1 = -2A \left((A - 1) + (A + 1) \cos \omega_0 \right)$$

$$b_2 = A \left((A + 1) + (A - 1) \cos \omega_0 - 2\sqrt{A}\alpha \right)$$

$$a_0 = (A + 1) - (A - 1) \cos \omega_0 + 2\sqrt{A}\alpha$$

$$a_1 = 2 \left((A - 1) - (A + 1) \cos \omega_0 \right)$$

$$a_2 = (A + 1) - (A - 1) \cos \omega_0 - 2\sqrt{A}\alpha$$

Reversing the high-shelf coefficients

Here, we can cheat because we from the diagram that $G = 4$. Everything becomes much easier if we know A .

To solve for ω_0 :

$$a_0 + a_2 = (A + 1)$$

$$-(A - 1) \cos \omega_0$$

$$+ 2\sqrt{A}\alpha$$

$$+ (A + 1)$$

$$-(A - 1) \cos \omega_0$$

$$- 2\sqrt{A}\alpha$$

$$a_0 + a_2 = 2((A + 1) - (A - 1) \cos \omega_0)$$

$$a_0 (1 + A_2) = 2((A + 1) - (A - 1) \cos \omega_0)$$

$$a_0 = \frac{2((A + 1) - (A - 1) \cos \omega_0)}{1 + A_2}$$

From earlier:

$$a_0 = \frac{a_1}{A_1} \text{ and } a_1 = 2((A - 1) - (A + 1) \cos \omega_0)$$

$$\frac{2((A - 1) - (A + 1) \cos \omega_0)}{A_1} = \frac{2((A + 1) - (A - 1) \cos \omega_0)}{1 + A_2}$$

$$\frac{(A - 1) - (A + 1) \cos \omega_0}{A_1} = \frac{(A + 1) - (A - 1) \cos \omega_0}{1 + A_2}$$

$$\begin{aligned} & A_1 (A + 1) - A_1 (A - 1) \cos \omega_0 \\ &= (1 + A_2)(A - 1) - (1 + A_2)(A + 1) \cos \omega_0 \end{aligned}$$

$$\begin{aligned} & (1 + A_2)(A + 1) \cos \omega_0 - A_1 (A - 1) \cos \omega_0 \\ &= (1 + A_2)(A - 1) - A_1 (A + 1) \end{aligned}$$

$$\cos \omega_0 ((1 + A_2)(A + 1) - A_1(A - 1)) = (1 + A_2)(A - 1) - A_1(A + 1)$$

$$\cos \omega_0 = \frac{(1 + A_2)(A - 1) - A_1(A + 1)}{(1 + A_2)(A + 1) - A_1(A - 1)}$$

$$\omega_0 = \arccos \frac{(1 + A_2)(A - 1) - A_1(A + 1)}{(1 + A_2)(A + 1) - A_1(A - 1)}$$

Now, to solve for Q we first much solve for α :

$$a_0 - a_2 = (A + 1)$$

$$-(A - 1) \cos \omega_0$$

$$+ 2\sqrt{A}\alpha$$

$$-(A + 1)$$

$$+(A - 1) \cos \omega_0$$

$$+ 2\sqrt{A}\alpha$$

$$a_0 - a_2 = 4\sqrt{A}\alpha$$

$$a_0(1 - A_2) = 4\sqrt{A}\alpha$$

$$\alpha = \frac{a_0(1 - A_2)}{4\sqrt{A}}$$

$$\frac{a_0(1 - A_2)}{4\sqrt{A}} = \frac{\sin \omega_0}{2Q}$$

$$Q = \frac{2 \sin \omega_0 \sqrt{A}}{a_0(1 - A_2)}$$

And f_0 is really easy:

$$2\pi \frac{f_0}{F_s} = \omega_0$$

$$f_0 = \frac{\omega_0 F_s}{2\pi}$$

Stage 2 filter

Cookbook definition of a high-pass filter

$$b_0 = \frac{1 + \cos \omega_0}{2}$$

$$b_1 = -(1 + \cos \omega_0)$$

$$b_2 = \frac{1 + \cos \omega_0}{2}$$

$$a_0 = 1 + \alpha$$

$$a_1 = -2 \cos \omega_0$$

$$a_2 = 1 - \alpha$$

Reversing the low-pass coefficients

This filter should be a lot easier to reverse due to the relative simplicity of the a_0 , a_1 , and a_2 definitions.

Again, we start by solving for ω_0 in terms of one of our normalized coefficients.

$$\cos \omega_0 = -\frac{a_1}{2}$$

$$\cos \omega_0 = -\frac{A_1 a_0}{2}$$

$$a_0 + a_2 = 1 + \alpha + 1 - \alpha$$

$$a_0 + a_2 = 2$$

$$a_0 (1 + A_2) = 2$$

$$a_0 = \frac{2}{A_2 + 1}$$

Substitute:

$$\cos \omega_0 = -\frac{A_1}{2} \frac{2}{A_2 + 1}$$

$$\cos \omega_0 = -\frac{A_1}{A_2 + 1}$$

$$\omega_0 = \arccos -\frac{A_1}{A_2 + 1}$$

We solve for α reusing our existing definitions:

$$a_0 = 1 + \alpha$$

$$\frac{2}{A_2 + 1} = 1 + \alpha$$

$$\alpha = \frac{2}{A_2 + 1} - 1$$

$$\alpha = \frac{1 - A_2}{A_2 + 1}$$

And then Q :

$$\frac{1 - A_2}{A_2 + 1} = \frac{\sin \omega_0}{2Q}$$

$$Q = \frac{\sin \omega_0 (A_2 + 1)}{2(1 - A_2)}$$

And ω_0 is the same as done previously.